OpenADMET CYP Inhibition Blind Challenge

Overview

Team horse-pusher entered both tracks of the challenge. The challenge covered CYP1A2, CYP2C9, CYP2D6 and CYP3A4, and the blind test set had 750 compounds. The regression track required us to predict pIC50 for all four isoforms and the TDI track required us to classify time-dependent inhibition for CYP2D6 and CYP3A4.

Both tracks used pretrained molecular encoders, bagged and ensembled models, and a final calibration step chosen to suit each track's metric.

Methodology

Training

We used the challenge training set, plus ChEMBL CYP activity data with overlapping compounds removed. We also used a pseudo-labelled pretraining set of about 273k compounds from Novartis. Addition of other datasets such as qHTS and Veith sets did not improve predictive power in our hands. We used Butina splits to generate the held out tuning sets from training data. There was noticeable benefit in including auxiliary heads - so the TDI inhibition, pIC50 data, and log2fc were used. Addition of features such as sterimol, RDKit, and charge features was useful for some endpoints but not for all, resulting in a different model mix for each endpoint.

Regression track

We pretrained on literature and pseudo-label data, then fine-tuned on the challenge data. Each isoform uses a different base model, based on a mix of chemprop, custom GNNs, and chemprop and Monroe embeddings with tabular models such as TabPFN and TabICL. We also implemented nearest neighbour correction to account for the test matrix being a dense matrix of similar compounds rather than being more separated in chemical space like the training set.

TDI track

We generated a number of candidate models, including a tuned chemprop classifier with RDKit descriptors, a LightGBM on a ECFP4, RDKit2D, charge, lability, sterimol and predicted-pIC50

blocks, and TabICL/TabPFN on frozen chemprop, and Monroe embeddings. All models were combined by Caruana greedy selection.

Calibration

All entries were calibrated to the board results, with the inferred positive rate inferred from classification entries, and a shift/distribution width changes inferred from relationships between spearman rho, R^2, and ST-RAE. This resulted in linear recalibration for regression 2D6 and 2C9 models and a change in sensitivity for TDI models, minimizing absolute error without affecting ranking capability.